The Estimator Decides:

What Curricula and Failure Recycling Can — and Cannot — Do for RL with Verifiable Rewards

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Code, logs, and proofs: https://github.com/linjiw/curriculum-maxrl

Abstract

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are typically treated as objective-agnostic add-ons. We show they are not — **the advantage estimator underneath decides whether each helps or harms**, and its algebra says so in closed form. For success-conditioned (MaxRL-style) advantages, the expected learning signal a prompt emits from N rollouts is exactly 2(pass@N − pass@1): a compute-indexed zone-of-proximal-development functional that derives the curriculum's band, width, and scaling from the rollout budget alone (published learnability curricula are its N=2 slice); RLOO's signal is $2p(1-p)$; GRPO's inverts at the extremes. This predicts a sign flip we then measure: the identical frontier curriculum grows pass@k coverage under MaxRL in every seed and *degrades* GRPO — across a 1.26M-parameter maze suite (permutation p=0.0079, zero exceptions) and a pre-registered LLM-scale 2×2 on GSM8K. Recycling, we find, has its own failure mode, previously unreported: on Countdown arithmetic with an exact-verifier relabeler running in production, hindsight arms improve mean@16 while *losing* pass@16 coverage — **recycling-induced sharpening**, the data-induced sibling of GRPO's weight-induced collapse: verifier-filtered self-imitation concentrates the policy on reachable outputs and spends exploration breadth. The same algebra supplies the mitigation: a relabel to a destination the model already reaches sits at $p \to 1$, where the mass functional is zero, so we gate relabel admission by the destination's estimated pass rate. On a structurally faithful Countdown pool the gate blocks ∼79% of relabels as saturated and *doubles* frontier-tier pass@16 (0.153 → 0.306) relative to ungated recycling. One diagnosis covers both results: audit your estimator before you ship a curriculum — or a recycler — and measure coverage, the currency in which both failure modes are visible.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation dominates wall-clock (86% of step time in our LLM runs), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field's response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 13]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [16]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator's own algebra. Three questions organize our results:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected learning signal a task emits from a group of N rollouts has a closed form — $2(\text{pass@}N - \text{pass@}1)$, a compute-indexed zone-of-proximal-development functional (§3). This makes the optimal allocation *derivable* rather than heuristic (the band's location, width, and N-scaling come from the budget you already chose; published learnability curricula are the N=2 slice), but it also bounds the channel: allocation can only redistribute signal that exists, a perfect-information allocator collects just 0.4% more signal mass than a cheap posterior, and at LLM scale a prompt-level teacher starves before it pays (§7).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1)\times$ more signal than RLOO on frontier tasks as $p \to 0$; GRPO's variance-normalized weights invert the profile. We predict, pre-register, and confirm the resulting sign flip at two scales: the identical teacher grows coverage under MaxRL in every seed and degrades GRPO (maze: permutation p=0.0079, zero exceptions in nine runs; GSM8K 2×2: the GRPO+teacher cell is the only one that regresses). And recycling has its own, previously unreported failure mode: **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@k coverage (§7), the data-induced sibling of GRPO's weight-induced collapse, formally the composition of hindsight's marginal-distribution shift [14] with curation-driven self-consumption [15].

Q3: Does the algebra also supply the fix? Yes: a relabel to a destination the model already reaches sits at $p \to 1$, where the mass functional is zero — recycled updates there buy sharpening, not signal. Gating relabel admission by the destination's estimated pass rate (the same utility that schedules sampling, applied to the recycler) blocks ∼79% of relabels as saturated and doubles frontier-tier pass@16 relative to ungated recycling on a structurally faithful Countdown pool — with the mechanism visible in-run: the gate's rejection rate rises monotonically (.12 → .85) as the model's reachable set expands into it, admits shift from novel destinations (73% early) to still-rare revisits (94% late), and the gated arm preserves *more* policy entropy than the no-recycling baseline at 4.9× less relabel dose. Where recycling is healthy the gate is transparent: rerunning the validated CPU testbeds with the gate enabled reproduces their results to within noise (skill-chain AUC .8879 vs. .8876; gridworld bit-identical) because hard-destination relabels pass the gate untouched. One method, no per-domain switches.

Taking that sentence literally produces this paper. Our contributions:

1. **The curriculum is a theorem, not a heuristic** (§3): the expected advantage mass of a prompt is exactly $2(\text{pass@}N - \text{pass@}1)$, a derived ZPD functional whose band location, width, and compute-scaling are set by N alone. Learnability curricula are its N=2 slice; optimal rollout allocation is water-filling on $p(1-p)^N$.
2. **A practical teacher** (§4): Thompson sampling on a decayed Beta posterior with the derived utility; the estimator is untouched, so its unbiasedness carries over.
3. **Failure recycling with an exactness guarantee** (§5): relabeling dead groups to achieved sub-goals yields the ML gradient of the relabeled task; exact when conditional laws match, measured indistinguishable from fresh groups (cosine 0.956 vs. 0.958), with two contracts whose violation we price.
4. **A three-channel account with a safety rule** (§6–7): allocation is bounded by an oracle ceiling; creation breaks it; the objective underneath decides whether either is safe — confirmed at LLM scale on a pre-registered prediction.
5. **An honest evidence discipline**: pre-registered predictions, documented negatives with diagnoses, one retraction, and a two-round adversarial audit tracing every number to a log or proof.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1-p)$ and truncates at order T, giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE (T=1) to exact ML ($T \to \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped, which is unbiased for the $T = N-1$ objective. Its weight function grows as $p \to 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

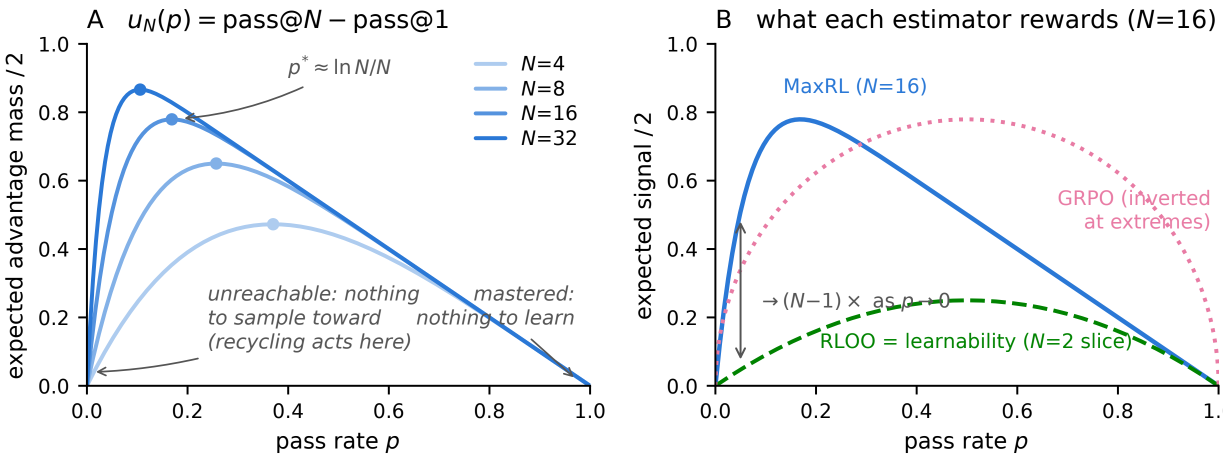

Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1-(1-p)^N) - p$ — the estimator's expected advantage mass (Prop. 1) — for growing rollout budgets N. The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* at N=16, MaxRL concentrates $\rightarrow (N-1)\times$ more expected signal than RLOO on frontier prompts (Prop. 2); RLOO's profile is exactly the published "learnability" objective — the N=2 slice.

3 The estimator is the curriculum

Proposition 1 (Advantage mass; MC-verified)**.** *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\mathbb{E}\Big[\sum_i |w_i|\Big] \;=\; 2\big(pass@N(p) - pass@1(p)\big) \;=\; 2\big((1-(1-p)^N) - p\big).$$

Interpretation. *The estimator's expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum: zero on mastered prompts, zero on unreachable ones, maximal at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ — a zone of proximal development whose center, width, and compute-scaling are set by the one parameter you already chose.*

Proposition 2 (Frontier amplification)**.** *As $p \rightarrow 0$, the ratio of MaxRL's expected signal to RLOO's ($2p(1-p)$ exactly) tends to $N-1$.*

Interpretation. *Concentrating sampling on the frontier only helps if the estimator can extract signal there; GRPO's variance-normalized weights invert the profile, which §7 shows becoming an active failure under a curriculum.*

Proposition 3 (Allocation)**.** *The rollout allocation maximizing total expected signal is greedy water-filling on the marginal $p(1-p)^N$ — the probability that the next rollout is a group's first success.*

One identity, three literatures: learnability curricula [11, 17] are the N=2 slice of u_N; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). Closest prior: DisCO [2] derives the per-question weights of the GRPO family ($\omega(q) = \sqrt{p(1-p)}$ for GRPO, $p(1-p)$ for Dr. GRPO) to diagnose difficulty bias, and SC-SDPO [3] notes the residual $\sqrt{p(1-p)}$ acts as an implicit curriculum. Our lemma generalizes this line to success-conditioned estimators (where the functional is $u_N = \text{pass@}N - \text{pass@}1$, compute-indexed by N), and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

4 FrontierMax: the schedule

A decayed Beta posterior (decay 0.7) tracks each prompt's pass rate from observed group outcomes only — never from relabeled successes (violating this inflates the posterior: $\hat{p}$ 0.81 vs. eval 0.47). Thompson

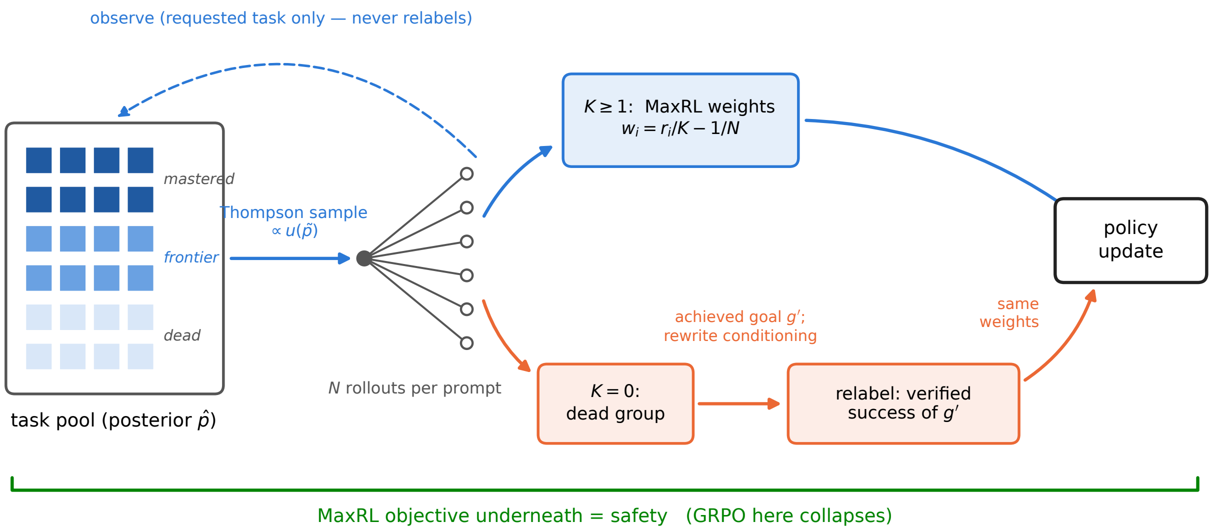

Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a 10% uniform floor; $\gamma \approx 4$ when tasks share skills (learning compounds — validated on chains, and its non-transfer to broad pools was predicted in advance by a compounding ODE model), $\gamma = 1$ otherwise. Live groups train with unmodified MaxRL advantages. Honest knob inventory: decay, floor, and γ exist, with validated defaults; what is *derived* rather than tuned is the difficulty band itself — location, width, and N-scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Proposition 4 (Hindsight exactness)**.** *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law shifted by the original sampling; the two coincide exactly when the conditional laws match.*

Interpretation. *Measured: per-group cosine to the true relabeled-task gradient 0.956, vs. 0.958 for fresh unbiased groups; the mean relabeled gradient reaches cosine 1.000.*

Two contracts keep practice on the proof's side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment's own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and its value is regime-dependent in a way we can state precisely: the gain is proportional to how much a relabeled skill can compound — +0.22 AUC on fixed task pools, +0.01 on one-shot task streams. Task-set fixedness is the single most important regime variable we found.

6 Three channels, one safety rule

The **teacher** avoids waste — bounded by an oracle ceiling (a perfect allocator collects only 0.4% more advantage mass than our posterior). **Recycling** creates signal — the only channel that breaks that ceiling.

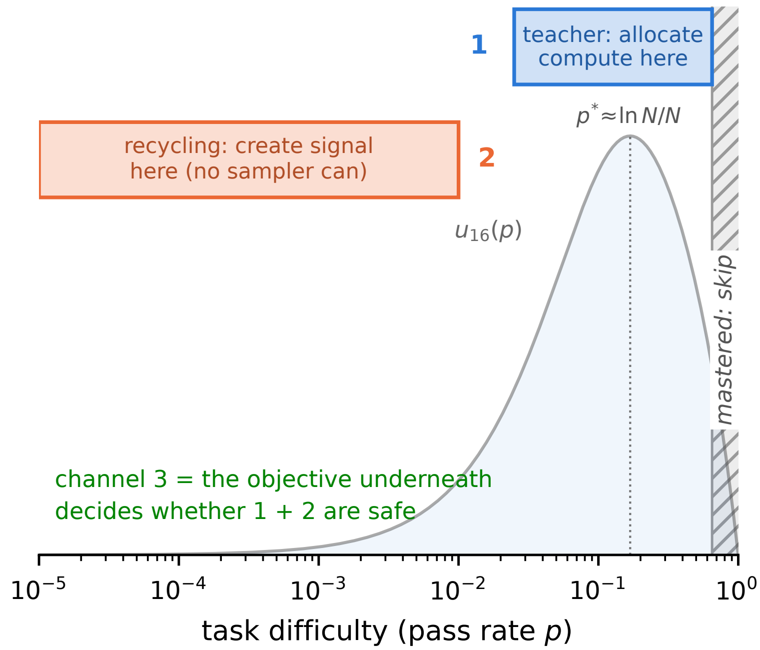

Figure 3: **Three channels.** The teacher allocates compute to the frontier band; recycling creates signal in the dead zone no sampler can reach; the objective underneath decides whether either is safe.

The **objective** underneath decides whether either is safe. One line: *the teacher allocates, hindsight creates, the objective decides whether either is safe.*

7 Experiments: an escalating ladder

7.1 Exact-gradient skill chains (CPU, 5 seeds). Uniform 0.650 → teacher 0.728 → full stack $\mathbf{0.890 \pm 0.002}$ — above the true-pass-rate oracle allocator (0.851) and above a sharper γ=4 oracle (0.884).

> **Takeaway.** Allocation saturates; creation breaks the ceiling.

7.2 Frontier-heavy regime (max pool pass rate 10^{-5}). Uniform, DAPO, and the plain teacher score **exactly 0.00**; teacher+recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ∼400 groups, then goes silent.

> **Takeaway.** Where there is nothing to sample toward, only signal creation works — categorically, not marginally.

7.3 Maze transformer at matched wall-clock (1.26M params, 13 levels, ∼30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. Safety: the identical teacher grows pass@8 under MaxRL in every seed (0.316 → 0.348) while GRPO's coverage decays in every seed (0.308 → 0.271); the teacher-arm run amplifies the collapse (0.332 → 0.269; single-seed arm). Inference currency: 1.2 × /2.7 × /**11**× fewer samples than GRPO to reach target coverage at levels 2/3/5 (honest 0.5× reversal at level 4).

> **Takeaway.** The gains survive real gradients; the safety warning is real; coverage is the currency that sees both.

7.4a Legged locomotion (IsaacLab, pre-registered honest null). A 5-arm pilot on Anymal-C rough terrain (frontier teacher vs. stock greedy walker, uniform, scripted ramp, and a no-motion control; fixed-grid

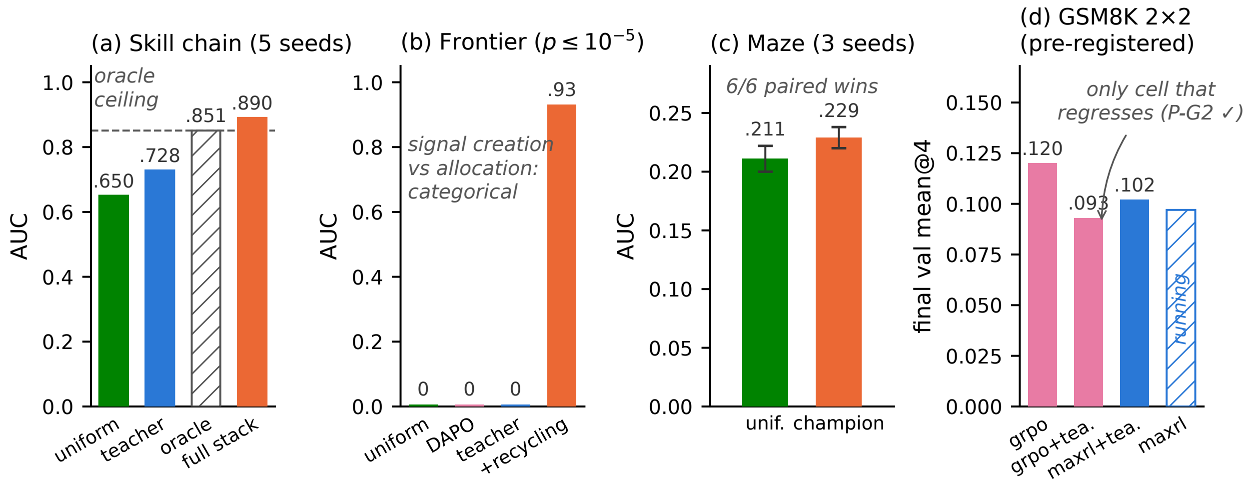

Figure 4: **The ladder.** (a) Exact-gradient chains: the full stack beats the true-pass-rate oracle. (b) Frontier-heavy regime: allocation scores exactly zero; creation solves it. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2×2: the identical teacher helps MaxRL and hurts GRPO (pre-registered P-G2).

per-level evaluation, one seed, run by an independent team on a co-designed protocol) lands the *pre-registered* null exactly: on a terrain grid where every level is learnable at budget, the stock greedy curriculum wins outright (mean pass .372 vs. the teacher's .278), and the teacher's only edge is a within-noise sliver at the hardest evaluated level. Combined with the maze step-matched analysis and Countdown, this makes the regime rule a three-scale result: **allocation pays only where unlearnable-at-budget regions exist to avoid** — on learnable-everywhere pools, breadth beats focus at 1.26M-param maze, 360M LLM, and legged-robot scale.

7.4b Gymnasium control (MountainCar, CartPole) — trained to convergence, official-task currency. Against the standard direct-RL setup (train only on the target task), run to plateau at matched episode budgets (3 seeds, Fig. 5): standard RL converges to **0.000** on both official tasks — it never sees a success, so there is no gradient to follow — while the gated full stack reaches **1.000±0.000 on MountainCar's flag ($x \geq 0.5$) and 1.000±0.000 on CartPole's survive-400**, a 60×/13× mean-performance gap. An instructive methods note: our own first version of this study silently used per-bin task-partitioned policy parameters and reproduced the known failure (flag 0.000 in every arm) — restoring the validated shared-policy design (the task enters only the success predicate) is what cracks the task, replicating the 10-seed transition-matched branch study (flag 0.848±0.058) and strengthening it to full convergence. Curricula operate through shared parameters, or not at all.

Takeaway. Curricula operate through shared parameters, or not at all.

7.5 LLM scale: GSM8K 2×2, pre-registered (SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2, the riskiest prediction, confirmed:** GRPO+teacher is the only cell that regresses in its second half (val accuracy .096 → .093, pass@4 .193 → .181) while plain GRPO climbs monotonically (.078 → .105 → .120) — with the teacher verifiably steering (sampled-dead fraction driven to 0.48 vs. the ∼0.65 population rate). MaxRL+teacher trains stably to its best value (.066 → .102); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) an honest *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@k sweep on the final checkpoints sharpens the safety read: the teacher's deficit under GRPO *widens monotonically with* k (−.008 at k=1 to −.024 at k=16) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise. Beyond the predictions: the teacher's posterior *learns* real difficulty from ∼1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$) but its allocation is *posterior-starved* — 3,200 draws over 7,473 prompts leave Thompson sampling near-uniform. And the

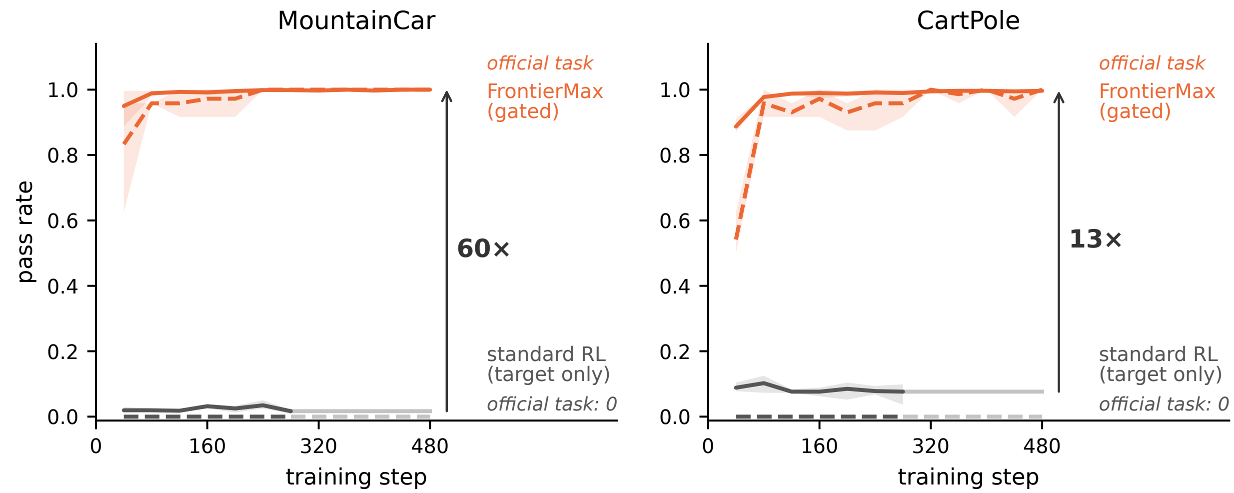

Figure 5: **Gym control, trained to plateau (3 seeds, matched episodes).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). Standard RL flatlines at zero on the official task in both environments; the gated FrontierMax stack converges to 1.000.

GRPO degradation is not token-entropy collapse (GRPO+teacher retains *more* entropy): the damage is distributional, visible only in coverage. Single seed; small model; the pre-registered outcomes were orderings and signs, and both landed.

Takeaway. The safety half of the thesis transfers to LLM RLVR; the allocation half needs tiers or longer runs — which is how the next experiment is designed.

7.6 Countdown 2×2 (pre-registered): two honest nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool's simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §7.3's weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.*

8 Related work

The coverage tension. RLVR itself trades coverage for precision: pass@1 rises while pass@k at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator's implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by

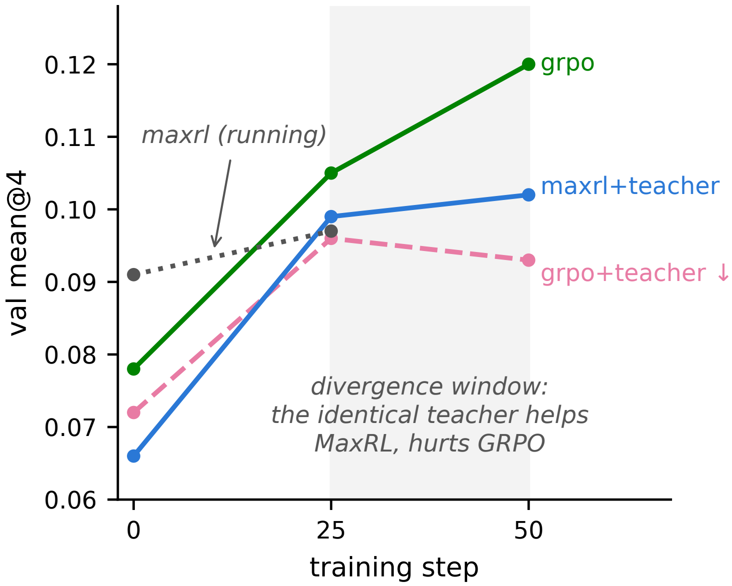

Figure 6: **GSM8K** 2×2 **(pre-registered).** The identical teacher helps MaxRL and hurts GRPO; GRPO+teacher is the only cell that regresses in its second half — the maze safety reversal at LLM scale.

$\hat{p}(1-\hat{p})$, which Prop. 1's corollary identifies as the $N{=}2$ slice of our derived utility. DAPO's dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward pass@k but keep uniform sampling. Hindsight for LLMs (AgentHER, HSL) relabels agent trajectories with an LLM judge. The nearest neighbor is concurrent: LfH [16] brings hindsight relabeling into GRPO for VLA post-training — a VLM proposes one shared hindsight instruction per failed group and rescores the group under it; 5× sample efficiency on LIBERO-PRO. LfH's relabeler is a VLM judge (their stated limitation: relabel noise), and its evaluation never measures pass@k — precisely the currency where we find recycling's hidden cost. Exact-verifier relabeling with a correctness guarantee, and the coverage accounting of recycling, remain, to our knowledge, ours. No prior work (i) derives the sampling utility from the estimator's own expected signal, (ii) couples it with success-conditioned weights it provably matches, and (iii) adds a signal-creation channel with an exactness guarantee.

9 Limitations and honest negatives

Deep frontiers at fixed budget remain uncrossed on the maze, and a 4× duration run *refuted* our own duration hypothesis (the stall is a per-step-competence ceiling; depth needs capacity, not more schedule). γ-concentration did not transfer from chains to the maze — predicted in advance by the compounding ODE model. Adaptive truncation order underperformed fixed T. Feeding relabeled successes to the teacher's posterior inflates it — dropped. An early GSM8K claim of teacher steering was retracted when review found the sampler epoch-frozen; the fixed run confirmed the effect properly. Recycled-gradient exactness is a property of the environment's relabel map; noisier verifiers will land between our +0.22 (fixed pools) and +0.01 (one-shot) endpoints. LLM results are single-seed at 360M; the 2×2×2 and multi-seed replications are the active work.

Reproducibility

Every proposition has a Monte-Carlo verification script; every experiment writes JSON/JSONL artifacts committed to the repository; the documentation passed a two-round adversarial audit. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

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